The Missing Trace

A Real-Time Data Integration and Process Optimization CTF Challenge

CITM-501: Information Systems Infrastructure
Assignment 6.6 – Capture the Flag Activity
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Team size: 3–4 players | Target completion time: 45 minutes | Version 2

Challenge Objective

NorthBridge Health Supply is a fictional distributor of serialized medical products. Every pallet received at the warehouse is scanned, posted to the ERP system, and passed in real time through an integration layer to a cloud traceability platform. Regulators and hospital customers rely on that platform to prove where each serialized item has been.

At 09:30 the following ticket arrives at the service desk:

```
INC-20931  Priority: High  Opened: 09:30  Caller: Warehouse Operations Lead
"The traceability dashboard is showing 8 receipts missing since about 09:10. They are all in
 the ERP. My team has been resending and rescanning to try to push them through. Please fix
 this before the 11:00 hospital shipment cut-off."
```

Your team is the integration support group. Using only the evidence in this packet, determine which component failed, exactly when it started failing, how many unique business transactions actually need to be recovered, and which long-term design prevents a repeat.

How to Play as a Team

The evidence is split into four work streams. Assign one role per person (combine roles 3 and 4 for a team of three). Each role contributes one part of the flag, so no one can finish alone. Plan to work in parallel for the first 20 minutes, then regroup and compare findings.

Role	Primary Evidence	Delivers to the Team
1. Data Reconciliation Analyst	A (Transactions), G (Scanner & Ops Log)	Which missing receipts are unique, which are duplicates, and which never reached the integration layer
2. Log Analyst	B (Hub Log), C (Proxy Log), H (Change Calendar)	The exact time the root-cause component began failing, and what changed
3. Infrastructure Analyst	D (Connectivity Tests), E (Vendor Status)	Which network path and component is broken, and which are healthy
4. Solutions Architect	F (Architecture Options) + findings from roles 1–3	The option that prevents recurrence without creating new data-quality risk

Rules

- Use only the evidence included in this packet. No outside lookups are needed.
- The component that reports an error is not necessarily the component that caused it. Compare evidence across layers.
- A “missing” receipt on the dashboard is not automatically a failed integration message. Trace each one.
- Separate retries, manual resends, and rescans from unique business transactions.
- Your final answer is a single flag in the exact format shown at the end. It is checked character-for-character.
- Two optional hints are available at the end of the packet. Note the time if you use them.

Current System Flow

1	2	3	4	5	6
Warehouse Scanner	ERP Receiving	Integration Hub	Legacy App Proxy	Traceability Cloud (TraceCloud)	Operations Dashboard
Wi-Fi VLAN 30 → HTTPS	Posts receipt, creates dedupe key	Queues, retries (3×, 30 s)	Relays hub → cloud, port 8443	Vendor API, port 443	Reads TraceCloud data

The dedupe key is generated by ERP from the purchase order and the serial-number set on the pallet. Two transactions with the same dedupe key describe the same physical receipt.

Business Impact

Operations uses the traceability dashboard to confirm that serialized products were received and are available for downstream picking and shipping. Products that are not visible cannot ship. When receipts appear missing, staff tend to resend or rescan, which creates duplicate serial records in the cloud platform if those duplicates are later replayed. Duplicate serial records must be manually investigated and can trigger compliance holds.

Evidence A – Transaction Timeline (ERP + Dashboard export, 09:29)

Txn ID	ERP Time	ERP Status	PO	Dedupe Key	Hub Status	Cloud Ack	Dashboard
T1001	09:02:14	Posted	PO-5510	RX-88410	Forwarded	ACK-200	Visible
T1002	09:05:29	Posted	PO-5511	RX-88411	Forwarded	ACK-200	Visible
T1003	09:06:11	Posted	PO-5512	RX-88412	Parked	None	Missing
T1004	09:08:08	Posted	PO-5513	RX-88415	Parked	None	Missing
T1005	09:10:03	Rejected (VAL-112)	PO-5514	—	Not received	None	Missing
T1006	09:11:30	Posted	PO-5515	RX-88419	Parked	None	Missing
T1007	09:14:17	Posted	PO-5516	RX-88421	Parked	None	Missing
T1008	09:18:04	Posted	PO-5515	RX-88419	Parked	None	Missing
T1009	09:21:39	Posted	PO-5517	RX-88426	Parked	None	Missing
T1010	09:23:01	Posted	PO-5512	RX-88412	Parked	None	Missing
TEST-DIRECT	09:25:10	N/A	—	—	Health test	HTTP 200	N/A

VAL-112 = ERP validation error: required field LOT_NUMBER is blank. Rejected receipts are never sent to the hub.

Evidence B – Integration Hub Log

```
08:00:00  CONFIG  retry_max=3  retry_interval=30s  dedupe_check=OFF  parked_queue_alert=OFF
09:02:15  MSG T1001  key=RX-88410  route=legacy-proxy:8443  attempt=1  result=HTTP_200  ack=ACK-200
09:05:30  MSG T1002  key=RX-88411  route=legacy-proxy:8443  attempt=1  result=HTTP_200  ack=ACK-200
09:06:12  MSG T1003  key=RX-88412  route=legacy-proxy:8443  attempt=1  result=HTTP_504_GATEWAY_TIMEOUT
09:06:42  MSG T1003  key=RX-88412  route=legacy-proxy:8443  attempt=2  result=HTTP_504_GATEWAY_TIMEOUT
09:07:12  MSG T1003  key=RX-88412  route=legacy-proxy:8443  attempt=3  result=RETRY_EXHAUSTED  status=PARKED
09:08:09  MSG T1004  key=RX-88415  route=legacy-proxy:8443  attempts=3  result=RETRY_EXHAUSTED  status=PARKED
09:11:31  MSG T1006  key=RX-88419  route=legacy-proxy:8443  attempts=3  result=RETRY_EXHAUSTED  status=PARKED
09:14:18  MSG T1007  key=RX-88421  route=legacy-proxy:8443  attempts=3  result=RETRY_EXHAUSTED  status=PARKED
09:18:05  MSG T1008  key=RX-88419  source=MANUAL_RESEND  attempts=3  result=RETRY_EXHAUSTED  status=PARKED
09:21:40  MSG T1009  key=RX-88426  route=legacy-proxy:8443  attempts=3  result=RETRY_EXHAUSTED  status=PARKED
09:23:02  MSG T1010  key=RX-88412  source=ERP_RECEIPT  attempts=3  result=RETRY_EXHAUSTED  status=PARKED
09:25:10  TEST-DIRECT  route=tracecloud-api:443 (temporary rule FW-TMP-77)  result=HTTP_200  health=OK
```

Evidence C – Legacy App Proxy Log

```
08:52:10  CHG-4471  OS security patch applied; relay configuration reloaded
08:52:11  truststore=/opt/lap/certs rebuilt from baseline image  entries=112
08:55:47  graceful restart completed; existing upstream pool connections retained  max_age=600s
09:02:15  relay T1001  upstream=tracecloud-api:443  pooled_conn=7  status=200
09:05:30  relay T1002  upstream=tracecloud-api:443  pooled_conn=3  status=200
09:05:47  pool=tracecloud  all pooled connections reached max_age; opening new upstream connections
09:05:47  upstream TLS handshake failed: unable to get local issuer certificate (TraceCloud Issuing CA G2)
09:05:47  connector_pool=tracecloud  status=UNAVAILABLE
09:06:12  inbound session accepted from integration-hub
09:06:42  upstream connection not established before timeout (30s); returned 504 to client
```

```
09:08:09  upstream connection not established before timeout (30s); returned 504 to client
09:11:31  upstream connection not established before timeout (30s); returned 504 to client
   ...    (same pattern repeats for every relay request through 09:23:02)
```

Evidence D – Connectivity Tests (Network Team, 09:25–09:28)

Source	Destination	Test	Result
Warehouse scanner (VLAN 30)	ERP Receiving :443	HTTPS	Success
ERP server	Integration Hub :443	TCP	Success
Integration Hub	Legacy App Proxy :8443	TCP	Success (4 ms)
Integration Hub	Legacy App Proxy :8443	POST /relay/tracecloud	HTTP 504 after 30 s
Legacy App Proxy	TraceCloud API :443	TCP	Success (22 ms)
Legacy App Proxy	TraceCloud API :443	TLS handshake	Failed – certificate verify failed (unknown issuer)
Integration Hub	TraceCloud API :443	HTTPS health check via FW-TMP-77	HTTP 200 – certificate chain valid (Issuing CA G2)
User workstation	Operations Dashboard	HTTPS	Success

Evidence E – External Platform Status (TraceCloud Status Page, 09:27)

All systems operational. No active incident. API health endpoint responding normally. No maintenance window in progress. Inbound processing normal for all regions during the last 24 hours.

Notice (posted Sep 28): TraceCloud completed rotation of its intermediate certificate to “TraceCloud Issuing CA G2.” Customers who maintain a local truststore or certificate pinning must import the G2 intermediate. Standard public trust stores already include it.

Evidence F – Architecture Options

- **Option A –** Keep the hub → legacy proxy → cloud path. Increase the hub timeout from 30 s to 120 s and retries from 3 to 10 so temporary upstream delays clear on their own.
- **Option B –** Point the ERP directly at the TraceCloud API and decommission both the integration hub and the legacy proxy, reducing the number of components that can fail.
- **Option C –** Keep the hub as the controlled integration layer, retire the legacy proxy hop with an approved direct TLS route from hub to cloud, enforce idempotency on the dedupe key, and alert on retry exhaustion and parked-queue age.
- **Option D –** Keep the hub as the controlled integration layer, retire the legacy proxy hop with an approved direct TLS route from hub to cloud, and add a scheduled job that automatically replays every parked message every 15 minutes.

Evidence G – Warehouse Scanner & Operations Log

```
09:06:05  WH-SCN-02  user=wh.recv1  PO-5512  pallet scanned  serials=48  lot=L24-0912
09:10:01  WH-SCN-04  user=wh.recv3  PO-5514  pallet scanned  serials=36  lot=<blank>  label damaged
09:11:24  WH-SCN-02  user=wh.recv1  PO-5515  pallet scanned  serials=60  lot=L24-0915
09:17:40  OPS-DESK   user=wh.ops2   "PO-5515 not on dashboard - used ERP Resend button"
09:20:55  WH-SCN-04  user=wh.recv3  PO-5512  pallet RESCANNED  serials=48  lot=L24-0912
                     note="not visible on dashboard, scanning again"
09:24:30  OPS-DESK   user=wh.ops2   "Told team to stop rescanning until IT responds"
```

Evidence H – Change Calendar (Today)

Change	Window	System	Description	Status
CHG-4460	08:30–08:40	Firewall	Rule update on warehouse wireless VLAN 30 (scanner network)	Completed
CHG-4471	08:50–09:00	Legacy App Proxy	Monthly OS security patch and config reload. Risk: Low. “No user impact expected.”	Completed
CHG-4475	22:00–23:00	ERP	Monthly database index maintenance	Scheduled
INC-20410	Sep 28	Legacy App Proxy	Manually imported TraceCloud Issuing CA G2 into proxy truststore after vendor rotation	Closed

Your Tasks

1. **Root cause.** Identify the component that is the root cause of the failure (use the component codes below).
2. **Onset.** Determine the time (HH:MM, 24-hour) at which the root-cause component itself first became unable to deliver data.
3. **Recovery scope.** Determine how many unique business transactions must be replayed through the integration layer. Do not count resends, rescans, or receipts that never reached the hub.
4. **Immediate recovery.** Describe the safest immediate recovery action, including what to do with the duplicate and non-integration items.
5. **Long-term design.** Choose the architecture option (A, B, C, or D) that best prevents recurrence while preserving real-time processing and data integrity.
6. **Control.** State one monitoring or control improvement that would have detected this problem before operations did.

Component Codes

Component	Code
Warehouse Scanner	SCANNER
Firewall	FIREWALL
ERP Receiving	ERP
Integration Hub	HUB
Legacy App Proxy	LEGACYPROXY
Traceability Cloud	TRACECLOUD
Operations Dashboard	DASHBOARD

Final Flag Format

`FLAG{<COMPONENT_CODE>_<ONSET_HHMM>_<UNIQUE_REPLAY_COUNT>_<OPTION_LETTER>}`

Formatting rules: uppercase, no spaces, no colon in the time. Example of the format only (not the answer): FLAG{ERP_1342_3_A}.

Team Answer Sheet

Submit the flag for verification. Complete the reasoning below so the playtest reviewer can confirm the flag came from analysis rather than guessing.

Team members and roles:

_____

1. Root-cause component and the two strongest pieces of evidence:

_____

_____

2. Onset time and the log line that proves it:

_____

3. Transactions to replay (list the IDs) and transactions excluded (with reason):

_____

_____

4. Immediate recovery action:

_____

_____

5. Architecture option and why the other three are weaker:

_____

_____

6. Monitoring / control improvement:

_____

FINAL FLAG:

_____

Start time / finish time / hints used:

_____

Optional Hints

Hint 1 (after 20 minutes): Line up the Dedupe Key column. Two different transaction IDs can describe the same physical pallet. Also check whether every missing receipt actually reached the hub.

Hint 2 (after 30 minutes): The hub records when it noticed the failure, not when the failure began. Find the earliest line where the failing component reports its own problem, and ask why it happened then rather than right after the restart.

References

CTF101. (n.d.). CTF 101. https://ctf101.org/

CTFd. (n.d.). What is Capture the Flag? https://ctfd.io/whats-a-ctf/